

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition? Answer Choices: A. It activates TLR5 on the surface of macrophages. B. It primarily induces the Th1-cell response. C. Peptidoglycan is its major antigen that induces an immune response. D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think> The patient is diagnosed with community-acquired pneumonia, and the treatment with cephalexin indicates that the likely causative pathogen is a bacterium, such as Streptococcus pneumoniae. The immune response elicited by bacterial pathogens typically involves the activation of the innate immune system, which recognizes pathogen-associated molecular patterns (PAMPs) through toll-like receptors (TLRs). TLR5 is specifically activated by bacterial flagellin, which is a component of the flagella of motile bacteria. While this is an important component of the immune response to some bacterial pathogens, it is not the most likely mechanism in this case. The Th1-cell response is an important aspect of the adaptive immune response to intracellular pathogens, such as viruses and some bacteria, but it is not the primary immune response elicited by the typical cause of community-acquired pneumonia. Peptidoglycan is a major antigen that induces an immune response in the case of bacterial infections, as it is a component of the bacterial cell wall. This is a key feature of the immune response to many bacterial pathogens, including Streptococcus pneumoniae. Encapsulation is a mechanism used by some pathogens, such as Streptococcus pneumoniae, to evade the immune response, but it is not the best description of the immune response elicited by the pathogen in this case. </think> <answer> C. Peptidoglycan is its major antigen that induces an immune response. </answer>

